Where Does Politeness Live in Hindi?
A Mechanistic Case Study of Honorific Encoding in Gemma Scope SAEs

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Abstract

Hindi encodes honorific level through pronouns (*aap*/*tum*/*tu*) and verb agreement. We ask whether Gemma-2-2B represents this level before any such marker appears, and whether public Gemma Scope SAEs expose a feature that captures it. We build a 600-pair contrastive Hindi set with a *context-only, pre-marker* condition: the input describes the social setting but contains no second-person pronoun or agreement-bearing form. On this condition, a linear probe on a middle layer reaches AUROC 0.722, against 0.610 for word TF–IDF, 0.550 for character n-grams, and 0.535 at layer 0. The signal holds with addressee roles held out (0.692) and on naturalistic prefixes (0.655). Because the contexts state the speaker–addressee relation, and the probe is binary (*aap* versus *tum*/*tu*), this shows that register-relevant social context is linearly decodable, not that the model holds a dedicated honorific variable.

The SAE analysis is negative. At layer 2, none of 50 candidates ranked by raw effect size passes our prespecified pronoun-specificity audit, and an exploratory five-prompt intervention on the top candidate shows no dose response or necessity. We therefore frame the work as a probing and negative-results study: it does not localise honorific information in the SAE dictionary or identify a causal handle. We release the dataset, rubric, and a single-GPU pipeline.

1 Introduction

Hindi honorific choice is both socially consequential and grammatically constrained. Addressing an elder with *tu* instead of *aap*, or pairing *aap* with a singular verb (*karta hai*) where plural-respectful agreement (*karte hain*) is required, produces a mismatch in register and agreement (Kachru, 2006). Whether multilingual language models represent honorific level internally, and where, is largely unknown.

Research questions. We ask three questions. **RQ1**: does a linear probe on Gemma-2 residual streams distinguish Hindi honorific level *when the pronoun and explicit agreement marker are absent*? **RQ2**: which Gemma Scope SAE features keep honorific specificity once both surface channels are audited? **RQ3**: does a tested candidate feature satisfy the causal triangle of correlation, steering sufficiency, and ablation necessity? A lighter, probe-only comparison with Japanese keigo and a native-speaker evaluation are reported in Section 4.4.

What this study can say about "where". The title asks where politeness lives, and our evidence answers only part of that. The layer-wise probe shows at which depth honorific-relevant context is most linearly accessible. The SAE search, by contrast, covers a single early layer, and the intervention covers a single feature. We therefore treat the paper as a probing and negative-results study: the SAE and intervention experiments show what a raw effect-size search does and does not recover, and they do not locate an honorific representation.

Motivation. Gemma Scope (Lieberum et al., 2024) provides pretrained SAEs at every Gemma-2 layer, which makes a targeted study feasible on one GPU. Probing work shows that multilingual models share cross-lingual structure (Pires et al., 2019) and that grammatical structure can be recovered from word representations (Hewitt and Manning, 2019). Whether social register is accessible in the same way has not been tested for Hindi.

Contributions.

1. A pre-marker probing result. Before any pronoun or explicit agreement marker is realised, a middle-layer probe reaches AUROC 0.722, against 0.610 for word TF–IDF, 0.550 for character n-grams, and 0.535 at layer 0. The result transfers to held-out addressee roles

(0.692) and naturalistic pre-marker prefixes (0.655). We state explicitly what it does not show (Section 4.1).

2. A prespecified SAE specificity audit with a negative outcome: none of 50 raw-effect-size-ranked candidates satisfies $A_{\text{hon}} \geq 0.80$ and $A_{\text{pron}} \leq 0.55$ on pronoun-stripped pairs.
3. An exploratory intervention on the top-ranked candidate, reported as a failure case rather than a causal test, since the audit shows that the candidate is a pronoun detector.
4. A 600-pair contrastive Hindi honorific dataset (CC-BY-SA) with a context-only condition for every pair, a 200-pair Japanese keigo analogue, a six-axis rubric rated by three native Hindi speakers on 192 generations, and a reproducible pipeline for a single 24 GB GPU.

2 Background and Related Work

Hindi honorifics. Hindi marks honorific register through two co-occurring grammatical mechanisms (Kachru, 2006). The first is the *second-person pronoun*: *aap* (formal and respectful, used with elders, strangers, and superiors), *tum* (familiar, used with peers), and *tu* (intimate or dismissive). The second is *verb agreement*: plural forms (*karte hain*, *gaye the*) obligatorily co-occur with *aap*, while singular forms (*karta hai*) co-occur with *tu* and typically with *tum*. The honorific level of an utterance is therefore encoded jointly by pronoun and verb agreement. Japanese contrasts with this: polite and plain verbal morphology (e.g., *-masu* versus the plain form) is a central realisation of politeness (Shibatani, 1990), and the keigo system also includes *sonkeigo* (referent-elevating) and *kenjōgo* (self-lowering) forms. Our Japanese comparison covers only the polite versus plain contrast.

Honorifics are not general formality. We use *honorific* strictly for the morphosyntactically encoded speaker–addressee relation. A formal news headline has no honorific level, and a casual request to a close friend has a low one. The contrastive pairs, confound controls (Section 3.2), and rubric (Section 3.4) are designed to isolate this distinction.

The pronoun-token confound. Because *aap*, *tum*, and *tu* are highly informative single tokens, a feature that fires on pronoun identity is not evidence of abstract honorific encoding. A pronoun-stripped probe addresses this risk at the probe level. Feature-level specificity needs separate, orientation-invariant honorific and pronoun AUCs, so we never infer specificity from effect size alone.

Sparse autoencoders. An SAE trained on a transformer residual stream learns an overcomplete dictionary of feature directions and decomposes each activation as a sparse positive combination (Templeton et al., 2024). Feature steering adds $+\alpha \mathbf{v}_f$, for decoder direction $\mathbf{v}_f$, to the residual stream at inference, and ablation subtracts it. Gemma Scope (Lieberum et al., 2024) releases pretrained JumpReLU SAEs at all layers of Gemma-2-2B and Gemma-2-9B.

Interpretability of multilingual and grammatical features. Interpretability work studies knowledge through probing (Alain and Bengio, 2017; Hewitt and Manning, 2019), causal analysis (Geiger et al., 2021), circuit discovery (Conmy et al., 2023), and representation editing (Meng et al., 2022). The closest multilingual work identifies, ablates, and steers with SAE-derived language-specific features (Deng et al., 2025); Zwirner et al. evaluate such steering across tasks and models (Zwirner et al., 2026). TenseLoC combines probing, causal tracing, and SAE steering for grammatical tense in eight languages, including Hindi (Tumurchuluun et al., 2025). Tense is a single-slot morphological choice tied to an objective time reference. Hindi honorific register instead requires coordinating a pronoun with matching verb agreement across a generation, and it depends on a socially conditioned judgment of addressee status. This is why it is unclear that a single decoded or steered direction could control it.

Honorifics in NLP. Prior work tests Korean mismatch surprisal (Noh et al., 2024) and sociopragmatic constraints (Cho and Kwon, 2026), and machine translation work controls politeness at the output level (Sennrich et al., 2016). We are not aware of earlier work that applies an orientation-invariant pronoun-confound audit to SAE features for an Indic honorific system. The closed pronoun set makes pronoun-token detection a concrete, checkable confound for raw effect-size ranking (Section 3.3), and the same audit could be reused for other closed-class address-term systems such as Japanese, Korean, or Javanese.

3 Experimental Setup

3.1 Data and Models

Models. We use Gemma-2-2B (≈5 GB in bf16; Gemma Scope JumpReLU SAEs at all 26 layers). The model plus one layer SAE peaks at 14.4 GB of VRAM in our runs, which fits on a single 24 GB RTX 4500 Ada. We use `transformer-lens` for residual-stream extraction (Nanda and Bloom, 2022) and `sae-lens` for SAE loading (Bloom et al., 2024).

Hindi contrastive pairs. We construct 600 minimally contrastive Hindi sentence pairs. The two members of a pair share propositional content and differ in honorific level (*aap* vs. *tum*, *aap* vs. *tu*, or *tum* vs. *tu*). Items span six domains: social, professional, family, public service, medical, and customer service. Each item comes with the following conditions (Table 1 illustrates the format):

- **Raw condition**: the completed sentence, including pronoun and agreeing verb.
- **Pronoun-stripped condition**: a variant with all second-person pronoun tokens removed, leaving verb agreement and context intact. This isolates honorific signal from pronoun-token identity.
- **Context-only, pre-marker condition**: a paired social scenario that specifies the speaker–addressee relation and the intended content, and ends before any realised target pronoun or agreement-bearing form. We extract the residual state at the final context token and ask whether it predicts the register of the next utterance. This is the primary condition for RQ1.

Members of a pair differ only in the honorific marker, so length and content are matched within a pair by construction. The first author, a native Hindi speaker, wrote the pairs from minimal-pair templates that vary only the honorific marker (the pronoun and the agreeing verb form). Honorific labels follow deterministically from the template, so no separate annotation pass was needed. The author checked naturalness on a held-out sample.

Japanese comparison set. The 200 pairs contrast *desu/masu* with plain forms and mirror the Hindi minimal-pair design: each pair differs only in polite versus plain verbal morphology.

Class	Pre-marker context (probe input)	Target form (not seen)
aap	*Speaker: junior employee. Addressee: department head. Intent: ask when the addressee arrives tomorrow.*	*aap kal kab aayenge?*
tum/tu	*Speaker: college student. Addressee: close friend. Intent: ask when the addressee arrives tomorrow.*	*tum kal kab aaoge? / tu kal kab aayega?*

Table 1: Schematic items showing the format of the context-only condition (romanised for readability). They were written to illustrate the format and are not verbatim dataset entries. The probe sees the context only, up to its final token; the target form is shown for reference.

3.2 Probing

For completed-sentence diagnostics we extract residual-stream activations at all 26 layers and mean-pool over token positions. For the context-only evaluation we instead take the residual state at the final context token, before generation begins. We train L2-regularised logistic regression probes ($C = 1.0$; Pedregosa et al. 2011) and report 5-fold cross-validated AUROC, grouped by pair identifier, on the raw, pronoun-stripped, and context-only sets. The primary claim rests on the context-only set, because its input ends before both direct morphological cues. We also evaluate a matched character n-gram baseline, a word TF–IDF baseline, a split that holds out addressee roles, and naturalistic prefixes that end before an overt marker.

Binary label. The probe is binary: *aap*-class versus *tum*/*tu*-class. This means that in a *tum* versus *tu* pair both members receive the same label, so such pairs carry no binary contrast. Our results therefore support claims about *aap* versus non-*aap* only. They say nothing about whether the model separates *tum* from *tu*, and a three-class or ordinal probe is left for future work.

Confound controls. Table 2 lists the controls we apply to separate honorific signal from surface confounds.

3.3 SAE Candidate Search and Audit

We use the Gemma Scope residual-stream SAE (16k features, JumpReLU, $\ell_0 = 53$) at layer 2, the SAE layer available for our feature-search pilot. This is an operational choice: layer 2 maximised the pronoun-stripped completed-sentence probe, whose layer profile is nearly flat. It is not evidence that layer 2 is a privileged or represen-

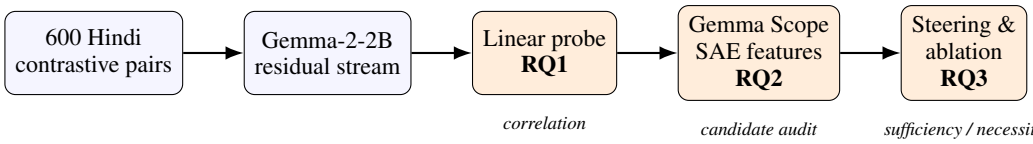


Figure 1: The analysis pipeline and the three legs of the causal triangle. Residual-stream activations from Gemma-2-2B feed a linear honorific probe (correlation), a Gemma Scope SAE candidate search (effect size plus a required orientation-invariant confound audit), and inference-time steering and ablation (sufficiency and necessity). Each stage maps to one research question.

Confound	Control applied
Pronoun token	Pronoun-stripped diagnostic plus pre-marker context-only probe for RQ1; prespecified $A_{\text{hon}}/A_{\text{pron}}$ audit for RQ2
Length & content	Minimal-pair construction: pair members differ only in the honorific marker
Pair leakage	Grouped CV blocks direct pair-member leakage; held-out-role evaluation tests a stronger social-context shift (Section 4.1)
Topic/register	Six domains (90–102 pairs per domain)
Verb-agreement suffix	Retained in the diagnostic pronoun-stripped set; absent from the primary pre-marker context-only input. Its contribution is a stated residual limitation of the completed-sentence feature audit

Table 2: Confound controls applied in this study. The pre-marker context-only condition and matched baselines address the concern that a probe merely detects overt pronouns or agreement morphology. They do not address social-role semantics (Section 4.1).

tative mechanistic locus, and it is not the layer at which the context-only probe performs best. The feature-level analysis is restricted to this SAE and layer. For each feature f we compute Cohen's d on the raw condition:

$$d_f = \frac{\mu_{aap}(f) - \mu_{tum/tu}(f)}{\sigma_{\text{pooled}}(f)} \tag{1}$$

Candidates are ranked by d_f. To test pronoun-independent specificity without directional ambiguity, we then compute two orientation-invariant quantities:

$$A_c(f) = \tfrac{1}{2} + \left|\text{AUC}_c(f) - \tfrac{1}{2}\right|, \tag{2}$$

with $c \in \{\text{hon}, \text{pron}\}$. Here A_{hon} is honorific discrimination on pronoun-stripped pairs and A_{pron} is raw-versus-stripped sentence discrimination. The prespecified passing rule is $A_{\text{hon}} \geq 0.80$ and $A_{\text{pron}} \leq 0.55$. Because ranking uses completed (raw) sentences, the audit tests whether this particular ranking recovers pronoun-independent candidates. It is not an exhaustive search of the SAE dictionary.

3.4 Intervention and Evaluation

We evaluate the three legs of the causal triangle (Geiger et al., 2021) for one candidate, the top-ranked feature f_{646}. The audit (Section 4.2) shows that this feature fails the specificity criterion, so the intervention should be read as a failure-case check on a pronoun-related feature. It is not a test of a validated honorific feature.

Correlation: probe AUROC and per-feature effect size above.

Sufficiency: inject $+\alpha \mathbf{v}_f$ at layer 2 on naturally *tum*-form prompts and measure the shift in addressee honorificity.

Necessity: inject $-\alpha \mathbf{v}_f$ at layer 2 on naturally *aap*-form prompts and measure the drop in honorificity. We also run encode-side zero-ablation, which zeroes the SAE encoding of feature f.

The decoder direction is unit-normalised before injection, so α is the magnitude added in residual-norm units, comparable across features and layers. Injection applies to generated positions only, which leaves the prompt context intact. We sweep $\alpha \in \{2, 4, 8, 16\}$.

Scope of the pilot. The sweep covers feature f_{646} at layer 2 on five informal prompts, with one generation per prompt at each α; ablations use five naturally formal prompts. This is an exploratory diagnostic. The sample is too small for stable confidence intervals or broad claims about generation.

Honorific proxy. Each generation is scored by a deterministic morphological proxy, $\text{hon}(x) = (n_{aap} - n_{tu})/n_{\text{tok}}$. Here n_{aap} counts *aap*-paradigm pronouns and their agreeing plural-respectful verb forms (*hain*, *kijiye*, *karte*, ...), n_{tu} counts *tu*-paradigm pronouns and singular agreement forms, and n_{tok} is the token count. The proxy ranges over $\sim[-1, 1]$, is easy to reproduce, and operationalises

Axis	What it measures	Scale
Addressee hon.	Pronoun appropriateness (*aap*/*tum*/*tu*)	0–3
Verb agreement	Plural verb with *aap*; singular with *tu*	0–3
Register consistency	Honorific level constant throughout	0–3
Context approp.	Pragmatic level for implied social context	0–3
Fluency	Independent fluency (control)	0–3
Content preserv.	Prompt faithfulness (control)	0–3

Table 3: Six-axis honorific rubric. The first four axes are honorific-specific; the last two are controls.

the addressee-honorificity axis of the rubric.

Human rubric. Table 3 shows the six-axis rubric used by the native-speaker raters. Separating addressee honorificity from verb-agreement consistency exposes a common failure mode in which the pronoun is correct but the agreement is not.

4 Results

4.1 RQ1: Contextual Honorific Signal Before Overt Marking

A high probe AUROC on completed sentences could simply reflect Hindi's overt plural-respectful agreement. We therefore treat the context-only, pre-marker condition as the primary test: its input ends before a second-person pronoun or explicit agreement marker is realised. Table 4 shows that a character n-gram classifier, a strong surface-form baseline, is close to chance (AUROC 0.550), as is the layer-0 probe (0.535). Word TF–IDF reaches 0.610. The best middle-layer L2 probe reaches 0.722, which is 0.187 AUROC above layer 0, 0.172 above character n-grams, and 0.112 above word TF–IDF.

The signal is not confined to one split. It reaches 0.692 when addressee roles are held out and 0.655 on naturalistic prefixes that end before an overt marker. These evaluations narrow the morphology-only account, but they do not isolate an honorific variable. The contexts state the speaker–addressee relation, so a model that merely represents social roles such as elder, friend, or supervisor would be expected to support a probe of this kind without any separate honorific representation. We did not measure how many items can be solved from role or scenario identity alone,

Representation or baseline	AUROC	Gain of best neural probe
Layer-0 representation	0.535	$+0.187$
Character n-gram	0.550	$+0.172$
Word TF–IDF	0.610	$+0.112$
Best transformer layer	**0.722**	–
Held-out addressee role (transformer)	0.692	–
Naturalistic pre-marker prefix (transformer)	0.655	–

Table 4: Pre-marker context-only probe controls. "Context-only" contains no realised target pronoun or agreement morphology. The gains are separate, explicit comparisons against each displayed baseline. The held-out-role and naturalistic-prefix evaluations test transfer beyond the templated marker configuration.

Candidate profile	d	A_{hon}	A_{pron}
f_{646} (raw rank 1)	2.209	0.514	0.808
All 50 candidates	–	≤ 0.518	0.656–0.808
Passing rule	–	≥ 0.80	≤ 0.55

Table 5: Prespecified two-way SAE specificity audit at layer 2. A_{hon} is honorific discrimination on pronoun-stripped pairs; A_{pron} is raw-versus-stripped sentence discrimination. No raw-ranked candidate passes.

and character n-gram and TF–IDF baselines are weak stand-ins for semantic role information. We therefore read the result as linear decodability of register-relevant social context. Section 5 lists the controls that would sharpen it.

4.2 RQ2: Raw-Ranked SAE Features Fail the Specificity Audit

We rank candidates by raw-condition Cohen's d and audit all 50 with the prespecified criterion: $A_{\text{hon}} \geq 0.80$ on pronoun-stripped pairs and $A_{\text{pron}} \leq 0.55$ for raw-versus-stripped sentence discrimination. No candidate passes. The top candidate, f_{646}, has a large raw effect size ($d = 2.209$) but $A_{\text{hon}} = 0.514$ and $A_{\text{pron}} = 0.808$. This is the failure mode one expects when pronoun identity is highly predictive: the ranking surfaces pronoun-related features, not a validated honorific feature. The result does not show that no such feature exists elsewhere in the dictionary, at another layer, or under a search conducted on pre-marker representations.

4.3 RQ3: An Exploratory Intervention on a Failed Candidate

Because f_{646} already fails the audit, this pilot is not a test of a validated honorific feature. It asks only whether the highest-effect-size candi-

Condition ($n = 5$)	Mean proxy	Δ vs. baseline
Baseline (informal)	-0.100	–
Steering, $\alpha = 2$	-0.091	$+0.009$
Steering, $\alpha = 4$	-0.091	$+0.009$
Steering, $\alpha = 8$	-0.072	$+0.028$
Steering, $\alpha = 16$	-0.216	-0.116
Negative ablation (formal prompts)	$+0.216$	no reduction
Zero ablation (formal prompts)	$+0.245$	no reduction

Table 6: Exploratory intervention pilot for f_{646} at layer 2. The automatic proxy is bounded approximately in $[-1, 1]$. Each condition contains one generation for each of five prompts; the results are descriptive, not inferential.

	Hindi	Japanese
Completed-sentence probe AUROC (best layer)	0.817	0.999
AUROC spread (26 layers)	0.021	0.001
Clear AUROC peak?	No	No

Table 7: Hindi vs. Japanese completed-sentence honorific probe results (descriptive only). The Hindi value is the diagnostic pronoun-stripped condition, not the primary 0.722 pre-marker result. Both languages are decoded almost uniformly across layers. The Japanese ceiling is consistent with the *desu/masu* suffix being a more regular surface marker than Hindi's pronoun+agreement pattern, but we did not test this explanation.

date moves honorific marking. We inject the unit-normalised decoder direction at layer 2, at generated positions only, on five informal prompts with one generation per prompt and dose. The morphology proxy is bounded approximately in $[-1, 1]$. Table 6 shows a non-monotonic dose response: the largest pre-collapse shift is only $+0.028$ at $\alpha = 8$, and $\alpha = 16$ reverses the direction. On five naturally formal prompts, both negative-direction and encode-side zero ablation preserve positive marking instead of reducing it. These results support neither sufficiency nor necessity, and the sample is too small for inferential statistics.

4.4 Supplementary Checks: Japanese Probe and Human Ratings

Japanese probe. This is a deliberately light check on how easily honorific labels can be read out in a second language. It is not a second causal study: we run only the completed-sentence probe on the Japanese set and repeat neither the pre-marker condition, the SAE audit, nor the intervention. Table 7 shows that Japanese *desu/masu* versus plain forms are decodable at AUROC ≈ 0.999, near ceiling, and that the curve is flat across all 26 layers (spread 0.001). The Hindi pronoun-stripped diagnostic reaches 0.817 with a spread of 0.021. Japanese sentences carry highly regular polite/plain suffixes, so near-ceiling decodability is expected and mainly shows that the marking is easy to read off. The two numbers are not directly comparable to the Hindi pre-marker result, and we draw no conclusion about cross-linguistic differences in internal mechanism.

Human ratings. Three native Hindi speakers, blinded to condition, rated 192 matched generations across eight conditions (24 prompts each) on the six 0–3 rubric axes (Table 3). This yields 576 generation–rater evaluations and 3,456 scalar judgments; per-condition ratings are in the supplementary material. Mean pairwise Spearman ρ was 0.79 for addressee honorificity, 0.72 for verb agreement, and 0.70 for register consistency. Four-shot prompting received the highest addressee-honorificity score (2.36 vs. 0.47 for the baseline). Top-20 steering stayed lower (1.31) and reduced verb agreement (2.01 vs. 2.68). These ratings agree with the automatic proxy and with the negative result for the additive interventions we tested. They cover only 24 prompts and do not establish population-scale pragmatic validity.

5 Discussion

The probe, SAE, and intervention results constrain the possible accounts, but they do not identify an internal mechanism.

What the probing result supports. Honorific-relevant information is linearly decodable from the residual stream before any pronoun or agreement marker appears, and it beats surface baselines and the layer-0 representation. This goes beyond simple marker recognition. The remaining alternative is contextual correlation: topic, lexical choice, syntax, and especially the explicitly stated social roles may predict honorific level without any dedicated honorific variable. The held-out-role and naturalistic-prefix results reduce this concern but do not remove it. Linear decodability also does not show that the model uses the decoded direction when it generates. The mechanistic conclusion we think the evidence supports is therefore modest: register-relevant social context is accessible to a linear readout in the middle of the network.

What we can and cannot say about "where". The probe indicates that context-only decodability is strongest at a middle layer. Our SAE search ran at layer 2, which was an operational choice, so the probing and SAE analyses are not aligned, and we cannot say which SAE features, if any, correspond to the probed signal. Moreover, candidates were ranked on completed sentences, although the main result concerns the pre-marker condition. This ranking favours features that respond to overt pronouns, and all 50 candidates failed the audit, with the top feature at $A_{\text{hon}} = 0.514$ versus $A_{\text{pron}} = 0.808$. The experiment therefore shows that raw effect-size ranking is confounded by pronoun identity. It does not test whether an honorific feature exists at the probe-maximising layers or under a search run directly on pre-marker representations.

Reading the intervention. The steering sweep is non-monotonic, and both ablations preserve formal marking. It uses one feature, one layer, five prompts, and one generation per prompt and dose, and the feature had already failed the specificity criterion. We treat it as a failure case that illustrates why a feature chosen by effect size should not be steered without an audit. It is not evidence for or against a causal honorific representation. Producing honorific Hindi also requires coordinated pronoun choice, verb agreement, and cross-clausal consistency, so a convincing test needs a larger design than ours.

What would settle the question. Four extensions would turn this study into a mechanistic one. (i) Three-class or ordinal probing, to cover *tum* versus *tu*. (ii) Counterfactual role reversals in which the content is fixed and only the social relation changes, with independently written contexts and a semantic role baseline, to separate honorific information from role semantics. (iii) An SAE search that ranks candidates on pre-marker representations, applies the same audit, and sweeps layers including the probe-maximising ones. (iv) Intervention on features that pass the audit, with more prompts, multiple decoding seeds, and confidence intervals. A pre-marker Japanese evaluation would also show whether the grammatical locus of marking changes what is decodable.

6 Conclusion

We studied how Gemma-2-2B and Gemma Scope SAEs represent Hindi honorific level. After a strict pre-marker control, a middle-layer probe reaches AUROC 0.722, above word TF–IDF (0.610), character n-grams (0.550), and layer 0 (0.535), with supporting held-out-role (0.692) and naturalistic-prefix (0.655) results. This shows that register-relevant social context is linearly decodable, but not that the model holds a dedicated honorific representation: the contexts state the social roles, and the probe is binary. None of 50 raw-ranked SAE candidates passes the pronoun-specificity audit, and a five-prompt pilot on the top candidate shows neither a monotonic steering response nor ablation necessity. We thus separate contextual decodability from SAE-feature validation and causal control, and we leave localisation open. We release the dataset, pre-marker controls, rubric, and analysis pipeline, which runs on a single 24 GB GPU.

Future work. Beyond the extensions in Section 5, natural next steps include Bengali, Tamil, and Marathi, graded register axes, instruction-tuned Gemma-2 variants, and a test of whether Japanese keigo's more local marking is easier to steer than Hindi's coordinated system.

Reproducibility. Code, the 600-pair dataset (CC-BY-SA), the rubric, per-layer probe results, and candidate feature indices accompany this paper as supplementary material and are available at `https://github.com/amethystani/lihaaz`.

Limitations

Task definition. The probe is binary (*aap* versus *tum*/*tu*), so *tum*-versus-*tu* pairs receive identical labels and carry no contrast. We make no claim about three-level or ordinal honorific structure.

Social-role confound. The context-only inputs state the speaker–addressee relation. A model that represents roles such as elder, friend, or supervisor, without a separate honorific variable, could produce the same result. We did not measure how many items are solvable from role or scenario identity alone, and the dataset is structured and template-based. The held-out-role and naturalistic-prefix evaluations are meaningful controls, not substitutes for counterfactual role reversals, independently written contexts, or open-domain validation.

SAE and layer alignment. The SAE analysis uses a single early layer (layer 2) that does not co-

incide with the best probe layer. The two-way audit covers 50 candidates ranked on raw sentences, and it cannot rule out other features, feature combinations, layers, or nonlinear mechanisms.

Intervention. The steering and ablation pilot covers one feature that had already failed the audit, one layer, five prompts per condition, and one generation per prompt. It does not support confidence intervals, multi-seed robustness, or claims about fluency and content preservation.

Scope. The primary probe, SAE, and intervention work uses Gemma-2-2B. A probe-only replication covers nine models (supplementary), but SAE and causal conclusions remain restricted to Gemma Scope. The Japanese comparison is probe-only and uses completed sentences with regular morphology. The native-speaker evaluation covers 24 prompts and eight conditions; it supports the automatic proxy on the tested rubric but not population-scale pragmatic validity. Grouped CV blocks pair mates, and the extra held-out-role and pre-marker evaluations improve generalisation evidence without settling it. Larger human studies, layer-swept intervention, instruction-tuned variants, and other Indic languages are untested.

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